

SCIENTIFIC REPORT · HEAD-TO-HEAD · BERTONE ET AL. 2026

eCTM outperformed hemoderivatives.

Head-to-head against autologous blood-derived products — the majority of the standard-of-care arm.

Background

OBJECTIVE

Autologous blood-derived orthobiologics — APS, PRP, ACS, PPP — are the dominant intra-articular biologic for equine osteoarthritis. In this pragmatic trial the Standard-of-care arm was clinician-selected; 64% of standard-of-care cases received a hemoderivative. A pre-specified subgroup analysis therefore compared eCTM head-to-head against the hemoderivative group.

Methods

COMPARATORS

Hemoderivative sub-cohort of clinician-selected blood-derived products (APS, PRP, ACS, PPP) vs. the full eCTM arm (n=76) — a single 1.5 mL intra-articular injection of cryopreserved umbilical protein/microparticle suspension. Same lameness (AAEP), pain and effusion scales, same follow-up (Weeks 0, 2, 4, 12), same blinded scoring.

ANALYSIS

Mixed-effects models with treatment × time as the fixed effect and horse as random. Pre-specified head-to-head contrast: eCTM vs. hemoderivatives. Reported p-values are two-sided and use the pooled standard-of-care variance estimate (Supplementary Figures S1–S3).

WHY IT MATTERS

Hemoderivatives require blood draw, chairside processing, and often multiple injections. Prior controlled evidence for blood-derived orthobiologics in equine osteoarthritis extends only to about 2 weeks. A single dose off-the-shelf allograft that beats hemoderivatives on lameness and pain has meaningful practice implications.

Interpretation

STRONGER THAN THE PREVAILING BIOLOGIC PARADIGM

In a pre-specified head-to-head contrast, a single dose of eCTM outperformed clinician-selected blood-derived orthobiologics on lameness (p<0.04 at Weeks 4 and 12) and pain (p<0.003).

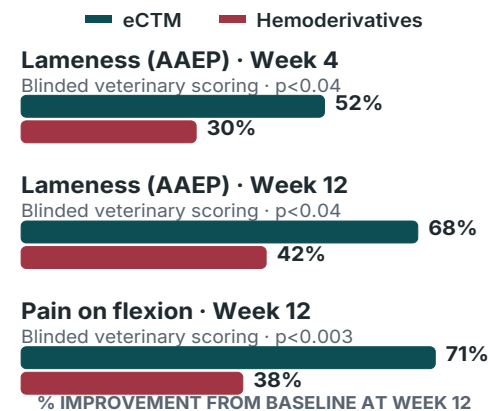
Results

HEAD-TO-HEAD vs. HEMODERIVATIVES

p < 0.003

Pain score at Week 12; lameness also favored eCTM at Weeks 4 & 12 (p<0.04).

FIG 1 · eCTM vs. hemoderivatives



eCTM produced greater improvement than hemoderivatives on lameness at Weeks 4 and 12 and on pain at Week 12 (Figure 8, Supplementary Figure S3).

Ready off-the-shelf. Stronger head-to-head.

One 1.5 mL injection of eCTM beat hemoderivatives on blinded lameness and pain endpoints.

Equine Performance Labs